Deveshwar Hariharan

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Education

Ph.D. and M.S., Electrical Engineering — **defended Mar 2026**
North Carolina State University — *Raleigh, NC, USA*
- GPA: 3.89/4.0

Bachelor of Technology, Electrical Engineering — **May 2018**
Amrita Vishwa Vidhyapeetham, Ettimadai Campus — *Coimbatore, Tamil Nadu, IN*
- GPA: 3.8/4.0

Skills

Expertise: Computer Vision, Machine Learning, Deep Learning, Cooperative Perception, Systems Engineering
Languages: Python, C++, MATLAB, CUDA
Libraries and Tools: ROS, TensorFlow, OpenCV, PyTorch, PCL, TensorRT, CARLA, Gazebo, KiCad
Management and Leadership: IRB Compliance, Cross-Functional Leadership, Advocacy, Governance

Experience

Data Science Research Consultant — **Aug 2025 - present**
Data Science Services - NCSU Libraries — *Raleigh, NC*
- Delivered data science consulting support to graduate students, faculty, staff, and external patrons on projects involving machine learning, data cleaning, automation, and analysis
- Served as a key resource for machine learning-focused requests, helping users implement AI tools
- Supported applied technology initiatives, including development work on a JetBot robotics project and ArcGIS credit usage management

Lead Graduate Research Assistant — **Jun 2019 – Aug 2025**
EcoPRT Autonomous Vehicle Lab, North Carolina State University — *Raleigh, NC*
- Lead electrical and computer science teams in autonomous vehicle development lab
- Develop and deploy object recognition algorithms to improve prediction accuracy and blindspot maneuvering using 3D point-clouds and cooperative perception
- Manage vehicle code to industry standards to improve runtime speed, code cross-compatibility, and to streamline research-to-test processes
- Design custom deployment PCBs to save the lab thousands of dollars in deployment costs for improved IMUs
- Lead 15 software engineering and robotics undergraduate students in developing a Fleet Management Server
- **Skills**: Computer Vision, Machine Learning, Cooperative Perception, ROS, C++, PCB Development, API Development, Cross-Functional Leadership

Student Researcher — **Jan 2019 – Jun 2019**
EcoPRT Autonomous Vehicle Lab, North Carolina State University — *Raleigh, NC*
- Developed the software for emergency braking upon detection of obstacles for autonomous vehicle
- Implemented a computer vision model and tracking algorithm to recognize obstacles
- Developed safety tests to pass IRB requirements to test autonomous vehicles on campus
- **Skills**: Computer Vision, Machine Learning, ROS, C++, Point Cloud Library (PCL), IRB Compliance

Community & Leadership

President of Graduate Student Body — **May 2021 – Apr 2023**
Graduate Student Association (GSA), NC State University — *Raleigh, NC*
- Created and funded Graduate Student Parent Association (GSPA)
- Collaborated to advocate for childcare and maternity/paternity benefits for graduate students
- Introduced and passed an amendment to the GSA Constitution to increase funding from GSA to Special Interest Groups (SIGs)
- Worked with SIGs to take advantage of the new amendment and increase funding
- Collaborated with peers from MIT and CMU in garnering support for the creation of the GRAD Caucus in the US House of Representatives
- Enlisted NC State University's GSA in the advisory team to the GRAD Caucus

Projects

Home Automation Projects | *Docker, Python, RaspberryPi, PlatformIO, ESPHome, Ollama* **Present**

- Build custom mmWave radars for presence and occupancy detection
- Deploy OSS tools for automation, logging, and control of IoT devices using Home Assistant
- Deploy fully local LLM-based voice assistants in multiple natural languages
- Build and integrate custom HiFi speakers
- Implement custom audio-sync protocol (Sendspin) to synchronize audio across several speakers

Automatic Cat Feeder | *Python, HTML, CSS* **Jun 2020**

- Solved the problem of being woken up by the cat early in the morning
- Built a custom cat feeder with automated feeding times, voice commands, and perception algorithm
- Now the cat goes to the feeder when he is hungry and it's feeding time, and we get to sleep

Line Following Robot | *Python, OpenCV* **Aug 2019 – Dec 2019**

- Developed a control system from scratch to maximize the time speed of the robot without deviating from the track

Lane Following drone | *ROS, Gazebo, MavROS, Deep Learning* **Jan 2019 – May 2019**

- Built and wired a drone from scratch
- Developed the ROS and Gazebo interfaces with MavROS for the flight controllers
- Developed and implemented a ROS node with a convolution neural network-based lane detector
- Developed the control systems for the vehicle to change course based on the lane predictions

Awards

Graduate Student of the Year, Leadership **2023**
Electrical and Computer Engineering Department, NC State University *Raleigh, NC*

Student of the Year, Academic **2015**
Electrical and Electronics Engineering, Amrita Vishwa Vidhyapeetham *Coimbatore, Tamil Nadu, India*

Publications

- **Hariharan, D.**, Zhu, Y., Pulithaya, D. U., Yoon, M., Hollar, S. (accepted) "RinneFormer - Cross Attention in Cooperative Perception", In review *Intelligent Vehicles Symposium, 2026*

- **Hariharan, D.**, Patil, N., Issa, J., Hollar, S., "Improvements to a Novel Autonomous Personal Rapid Transit (PRT) Prototype Vehicle", In *International Design Engineering Technical Conferences and Computers and Information in Engineering Conference, 2024*

- Chu, Y., **Hariharan, D.**, Hollar, S., Feng, F., "What Does That Car Mean? The Influence of Vehicle Motion and Symbolic Patterns of LED Signals on Pedestrians' Interpretation of a Vehicle's Intent", *Proceedings of the Human Factors and Ergonomics Society Annual Meeting, 2022*

- **Hariharan, D.**, PointPillars++, A Spatial Feature Encoder for Classification of Raw Point Cloud Data from LiDARs, *Electronic Thesis, NC State University 2022*

- **Hariharan, D.**, Venkat, G., Vasudevan, S. K., "A Novel Approach for increased transaction security with Biometrics and One Time Password A complete implementation.", *International Journal for Advanced Intelligence Paradigms, 2024*